🧬 NGS Hackathon Challenge

Master in Research and Innovation in Biotechnology — “Challenges in Biotechnology”

Date of the assignment: Thursday 30/01/2025

Date assignment is due: Friday 28/02/2025 – 23.59h

Marking: NGS Hackathon Challenge contributes 30% for the module

🎯 Introduction

In this module, you will act as a **bioinformatician** analyzing **Illumina next-generation sequencing (NGS)** data.
Your goal is not only to execute computational tools but to **understand, interpret, and improve** the quality of your analyses through evidence-based reasoning and iteration.

This assessment simulates a real-world bioinformatics scenario where reproducibility, accuracy, and interpretation are key.
Throughout the assignment, you will:

- Work with raw FASTQ files and identify quality issues.
- Use standard bioinformatics tools (`FastQC`, `Cutadapt`, `Bowtie2`, `Samtools`, `MultiQC`).
- Apply parameter optimization and critical thinking to improve results.
- Document everything in **Markdown** for reproducibility.

> 🧠 **Most important:** every section of your report must include a **strong biological interpretation** of the computational results.
> Explain what your observations mean in biological terms — e.g. sequencing quality, sample contamination, genomic complexity, or organismal biology.
> Interpretation will be the **most heavily weighted** criterion in grading.

📋 Assessment Overview

You will complete **three main tasks** (plus one extended exercise).
Each builds on the previous one, representing together **40 % of the module grade**.

Deliverables:

- Submit a **single PDF report** (generated from Markdown) to ADI before the deadline.
- Include all code, outputs (figures, tables, screenshots), and your biological interpretations.
- Demonstrate iterative improvement, clear reasoning, and teamwork.

🚀 What Is a Hackathon?

A **hackathon** is a collaborative and competitive event where participants solve technical problems in a short, intense timeframe.
In bioinformatics, hackathons are used to develop or benchmark workflows, test algorithms, and promote open collaboration with real datasets.

Relevant examples:

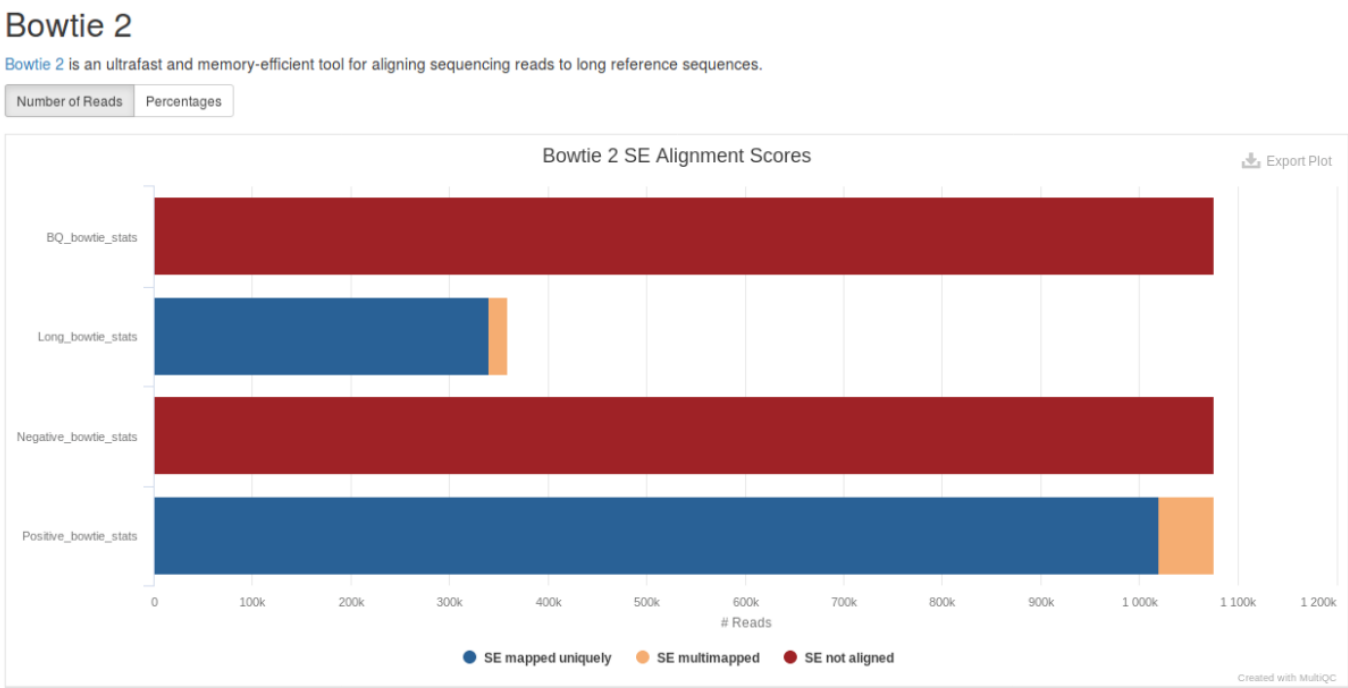
- **NCBI/NIH Bioinformatics Hackathons** — organized by the National Center for Biotechnology Information (NIH/NCBI) to prototype genomic analysis pipelines:
  🔗 https://ncbiinsights.ncbi.nlm.nih.gov/tag/bioinformatics-hackathon/
- **EMBL-EBI BioHackathon Europe** — focused on NGS interoperability and FAIR data standards:
  🔗 https://biohackathon-europe.org/

In this course, you will participate in a **mini-hackathon**, where teams will compete to achieve the **best mapping performance** (highest alignment rate and lowest error) on the same dataset.
Teamwork, creativity, reproducibility, and biological insight will determine success.

🧪 Assessment Tasks

Question 1 – Diagnosing and Re-mapping `Negative.fq`

You were given a FASTQ file (`Negative.fq`) from a simulated whole-genome sequencing experiment.
When reads were split using four barcodes (Negative, Positive, Long, BQ), the file `Negative.fq` showed **poor alignment**.



Your task

1. Identify **two plausible solution** for the poor mapping.
2. Test both hypotheses by applying corresponding corrections or parameter changes.
3. Re-map the reads against the reference genome.
4. Choose which hypothesis is most accurate, justify your choice, and provide a biological interpretation.

Hints

- Use `less` and `fastqc` to inspect raw reads.
- Try trimming options with `cutadapt` and different alignment modes with `bowtie2`.
- Evaluate performance with `samtools stats`, `samtools flagstat`, and `multiQC`.

Include in your report

- All commands used.
- Description of both initial hypotheses and solutions.
- Identification of the correct cause and justification with evidence.
- Table comparing both alignments (mapping %, error rate, properly paired reads).
- **Strong biological interpretation** of what these findings imply about your sequencing data and experiment.

Question 2 – Quality Trimming and Re-mapping `BQ.fq`

The dataset `BQ.fq` also produced low alignment rates.
Your objective is to **improve mapping accuracy and yield** through quality filtering and parameter optimization.

Steps

- Inspect raw quality and adapter content with `fastqc`.
- Use `cutadapt` to trim adapters, low-quality tails, and short reads.
- Re-map cleaned reads with `bowtie2`, testing different parameter sets.
- Compare **alignment rate**, **error rate**, and **number of properly paired reads** using `samtools` and `multiQC`.
- Reflect on the trade-off between **alignment rate** (quantity) and **alignment accuracy** (quality).

Include in your report

- All trimming and mapping commands.
- Tables with mapping statistics.
- Explanation of how trimming thresholds affected results.
- Discussion of which parameters produced biologically most meaningful results.
- **Biological interpretation:** e.g., does quality trimming remove contamination, low-quality regions, or artifacts from sample prep?

Question 3 – 🧠 Hackathon Challenge (“The Box”)

This is the **competitive hackathon** section.
You will use your optimized workflow from Question 2 to produce the **best possible mapping** for the `BQ` dataset.

Rules

- You may use either standard commands or your **own developed scripts or pipelines** (custom Bash scripts, or Python wrappers).
- Creativity and reproducibility are encouraged — clearly document your workflow so that others could reproduce your results.
- Each team must submit one final short **command or script BOX** and final statistics.

Deliverables

- A **BOX** (or script snippet) containing your final `cutadapt` and `bowtie2` commands or full custom pipeline.
- Final **mapping table** with alignment %, mismatches/error rate, and total reads.
- Short (≤ 300 words) explanation of your optimization strategy and how it improved mapping results.
- A **biological interpretation** of what your optimizations imply about the sample or sequencing process.

Example BOX

```
# Cutadapt trimming
cutadapt -q ¿¿¿¿¿ -m ¿¿¿¿¿ -a ¿¿¿¿¿ -A ¿¿¿¿¿¿¿??????

# Bowtie2 alignment
bowtie2 -x reference_genome -1 BQ_trimmed_R1.fastq -2 BQ_trimmed_R2.fastq
\
  ¿¿¿¿¿ -p ?????? -S BQ_aligned_tried?.sam
```

Evaluation criteria

1. Highest % of mapped reads.
2. Lowest mismatch/error rate.
3. Workflow clarity and reproducibility.
4. Quality of **biological interpretation**.

Question 4 – Exploring SAM Flags (Advanced Task)

Use **`samtools`** to split and inspect your alignment file (`BQ_aligned.sam` or `.bam`) based on mapping characteristics.

Tasks

a) Split into two files:

- Reads mapping to multiple locations (multimapping).
- Reads mapping uniquely.

b) Split into:

- Unmapped reads.
- Mapped reads.

c) Explain how you could achieve (b) directly using `bowtie2` parameters instead.

d) Use **BLAST** to identify the origin of unmapped reads and summarise findings. Report your results.

🧩 **Bonus (optional, extra marks)**
e) Split reads with **> 3 mismatches** vs **< 3 mismatches** and discuss whether these mismatches are biologically meaningful (e.g., real variants, errors, contamination).
f) Separate **trimmed** from **untrimmed** reads and interpret differences.

Hints

- Use `samtools view -f` and `-F` for flag filtering.
- Reference for flags: https://broadinstitute.github.io/picard/explain-flags.html
- Combine `grep`, `awk`, or regular expressions for more complex filters.

Include

- All filtering commands used.
- Screenshots or tables showing your outputs.
- Short answers to (c) and (d) with **biological interpretations**.

🗾 Submission and Markdown Instructions

Submit a **single PDF file** including:

- Code boxes and scripts.
- Figures and screenshots.
- Tables with summary statistics.
- Written answers with explicit biological interpretation.

🗺 Creating Your Report in Markdown

You can use Word or Google Docs, but **Markdown** is highly recommended for reproducibility.

🔹 **Option 1: Dillinger.io**

1. Open https://dillinger.io/.
2. Write your Markdown text directly.

3. Use the preview pane to visualise formatting.
4. Export or print to **PDF** or **HTML**.
5. Include figures as:
   `![description](figure.png)`.

🔹 **Option 2: Visual Studio Code (VSC) + Extensions**

1. Install Visual Studio Code.
2. Open the Extensions panel (`Ctrl + Shift + X`).
3. Install:
   - **Markdown All in One** → link
   - **Markdown PDF (yzane)** → link
4. Create a file named `NGS_Hackathon_Report.md`.
5. Write using Markdown syntax:
   - `#` for titles
   - `bash …` for code blocks
   - `![caption](image.png)` for figures
6. To export: `Ctrl + Shift + P` → *Markdown PDF: Export (pdf)*.
7. PDF will appear in the same folder.

🔌 Use `Ctrl + Shift + V` for live Markdown preview in VS Code.

✡️ Recommended Markdown Resources

- https://www.markdownguide.org/
- Markdown PDF (yzane)
- Markdown All in One